Annabel Li

Engineering Physics · University of British Columbia

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Education

B.A.Sc., Engineering Physics | *University of British Columbia* Expected May 2029

Technical Skills

Mechanical Design & Manufacturing	CAD (Autodesk Inventor, SolidWorks, OnShape); UBC PHAS Machine Shop certified (lathe, mill, waterjet); GD&T; DFM; CNC, welding, laser cutting, carbon-fibre layup; CFD (Star CCM+)
Embedded / Systems	ESP32, Arduino, servo control, PWM, embedded C/C++; Git, Linux
Software / ML	Python, C/C++, Java, MATLAB; NumPy, PyVista, TensorFlow, Keras; FPGA toolchain (Vitis, Vivado)

Experience

Mechanical & Research Co-op | *DarkVision Technologies Inc.* Jan. – May 2026

- Owned the end-to-end production of a long-term **corrosion test rig** from concept to build, integrating custom and externally sourced parts to meet manufacturing and mechanical constraints
- Designed test fixtures and mounts in **Autodesk Inventor** for experiments with medical-grade ultrasound hardware, fabricating components through laser cutting, welding, and CNC machining
- Built **MATLAB, Python, and PyVista** pipelines to analyze and visualize 2D ultrasound scan data
- Produced industry-standard technical drawings with **GD&T tolerancing** and led vendor communications for outsourced custom parts

Deep Learning Research Assistant | *ATLAS Canada / TRIUMF* May – Sept. 2025

- Selected via the Eric Vogt Summer Research Award to independently investigate neural-network optimization for the ATLAS Level-0 hardware trigger under high data-pileup conditions
- Compressed TensorFlow/Keras models for FPGA deployment, cutting inference time **20–30%** and model size **50–60%**
- Automated the training, evaluation, and debugging workflow in **Python**, increasing experiment throughput **4**× per day and eliminating manual reruns
- Implemented modular neural networks in **C++** with Vitis and Vivado, designing fixed-point data pipelines to satisfy hardware numerical constraints
- Presented results at TRIUMF Science Week, placing **3rd** among undergraduate and graduate presenters

Aerodynamics Engineer | *UBC Formula SAE* Sept. 2024 – Dec. 2025

- Spearheaded the end-to-end development of the beam wing for the 2025 competition vehicle, driving aerodynamic research, simulation, manufacturing, and design documentation
- Improved aerodynamic balance by **2–3%**, with drivers consistently reporting better handling in track testing
- Drove iterative CAD **(SolidWorks)** and CFD **(Star-CCM+)** development while incorporating feedback from teammates across vehicle subsystems, then manufactured the final component to tight tolerances via CNC machining and carbon-fibre layup

Projects

5-DOF Robotic Arm — Robot Summer Competition | *UBC Engineering Physics* May – Aug. 2026

- Designed and manufactured a 5-DOF robotic arm from scratch, owning mechanical architecture, actuator selection, fabrication, and mounting through to a competition-ready build within 4 weeks
- Developed the **ESP32 motion-control stack** in **C++**, including a custom PWM driver, multi-joint control, and higher-level task sequences; integrated the arm with the robot's state machine for autonomous task execution
- Unified mechanical, actuator, and software interfaces and debugged system-level issues through iterative hardware and firmware testing
- Built and served a browser-based control panel directly from the ESP32, providing live joint telemetry, manual jog controls, and sequence debugging

Honours & Awards

- **UBC Trek Excellence Scholarship** (2025, top 5% in faculty) · **Eric Vogt Summer Research Award** (2025) · **C. K. Choi Entrance Scholarship** (2024)